bldc_controller_routed

Design Report

Rev 1 | 2026-11-04 | jlcpcb

kicad-tools 0.27.0

Board Summary

Property	Value
Layers	4 copper (F.Cu, In1.Cu, In2.Cu, B.Cu)
Footprints	42 (36 SMD, 6 THT, 0 other)
Nets	37
Traces	697 segments
Vias	137
Board Size	70.0 x 90.0 mm

Stackup

Layer	Type	Thickness (mm)	Material
F.Mask	Top Solder Mask	0.01	--
F.Cu	copper	0.035	--
dielectric 1	prepreg	0.2	FR4
In1.Cu	copper	0.035	--
dielectric 2	core	1.0	FR4
In2.Cu	copper	0.035	--
dielectric 3	prepreg	0.2	FR4
B.Cu	copper	0.035	--
B.Mask	Bottom Solder Mask	0.01	--

Design Overview

Theory of Operation

Sensored BLDC Controller

Communication Interfaces

Protocol	Signals
SPI	MISO, MOSI, SCK

Assembly Notes

1 fine-pitch component; 2 thermal pads; 4 polarized components

- **Fine-pitch components**: 1 (U1)
- **Thermal pads**: 2 -- verify solder paste apertures
- **Polarized components**: 4 -- check orientation markings

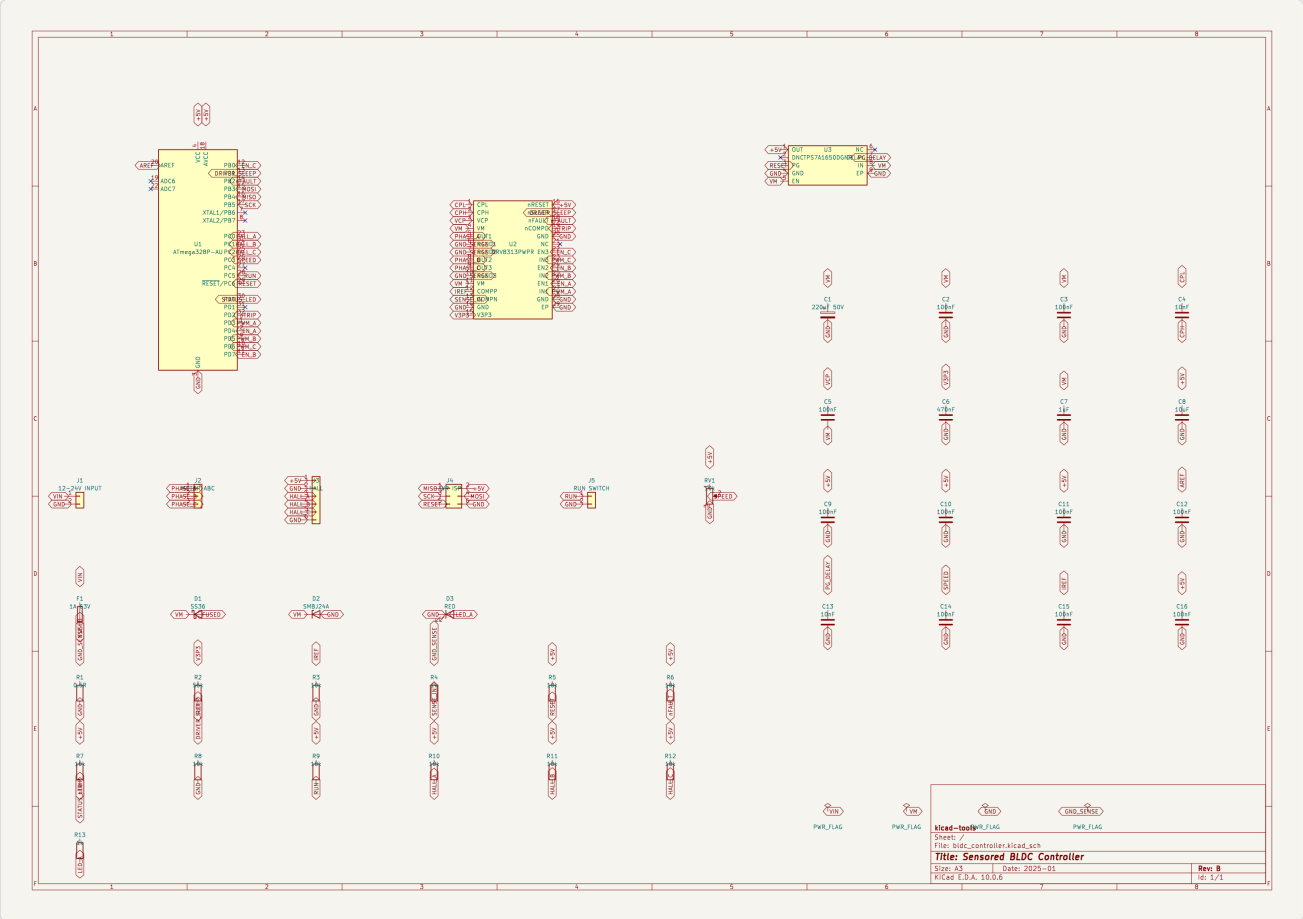
ERC Status

Metric	Count
Errors	0
Warnings	0

Status: PASS

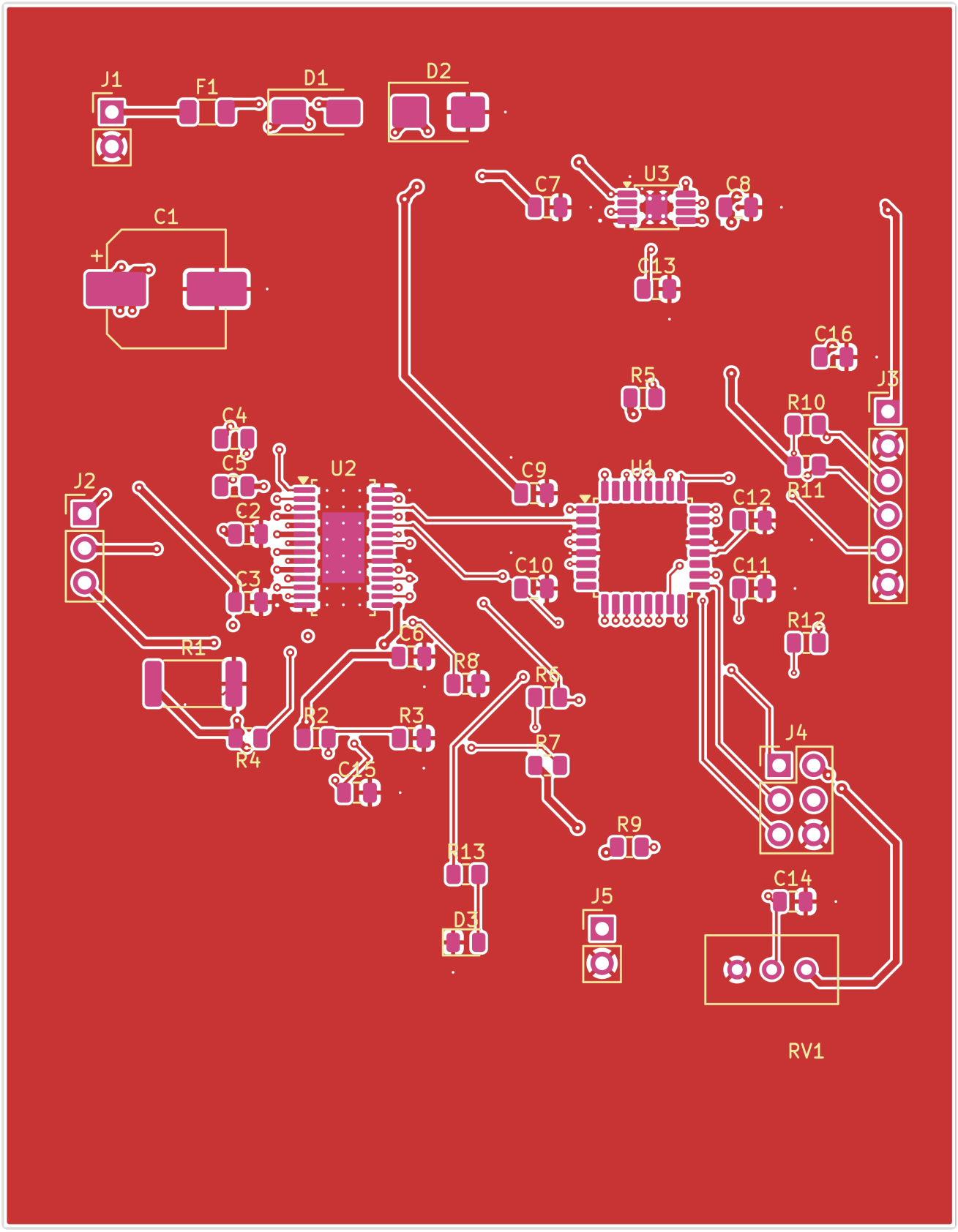
Schematic Overview

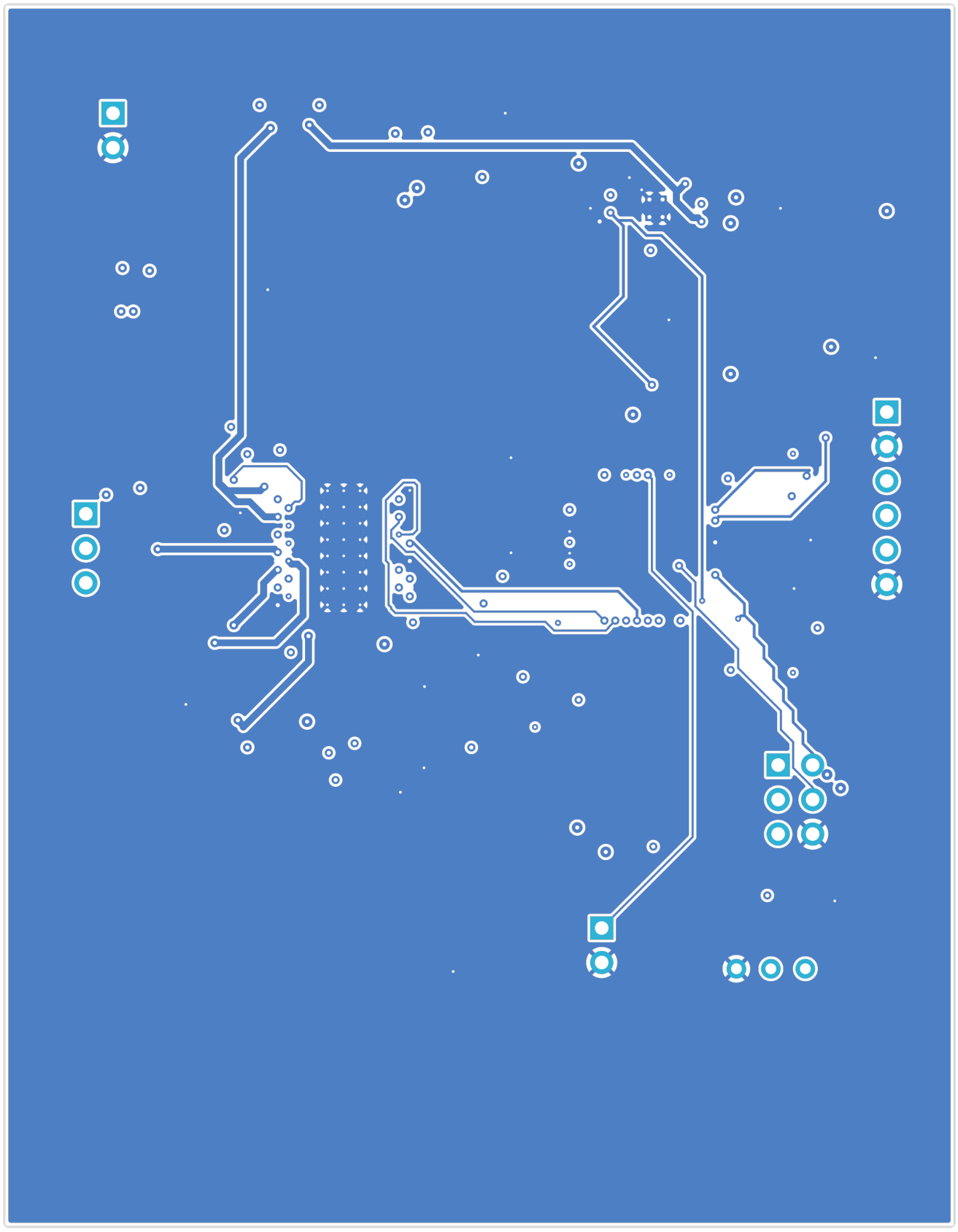
Schematic: bldc_controller



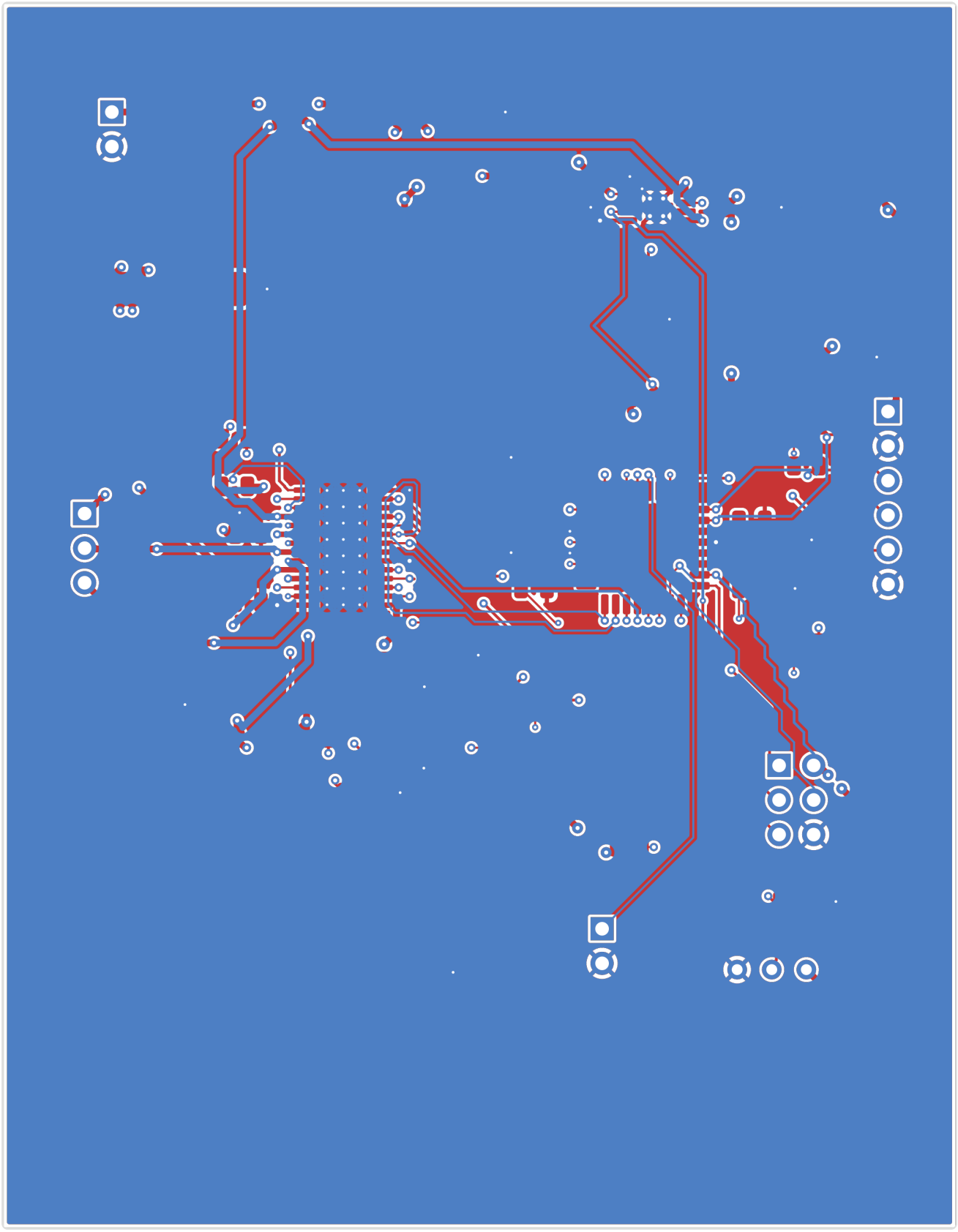
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PCB Layout

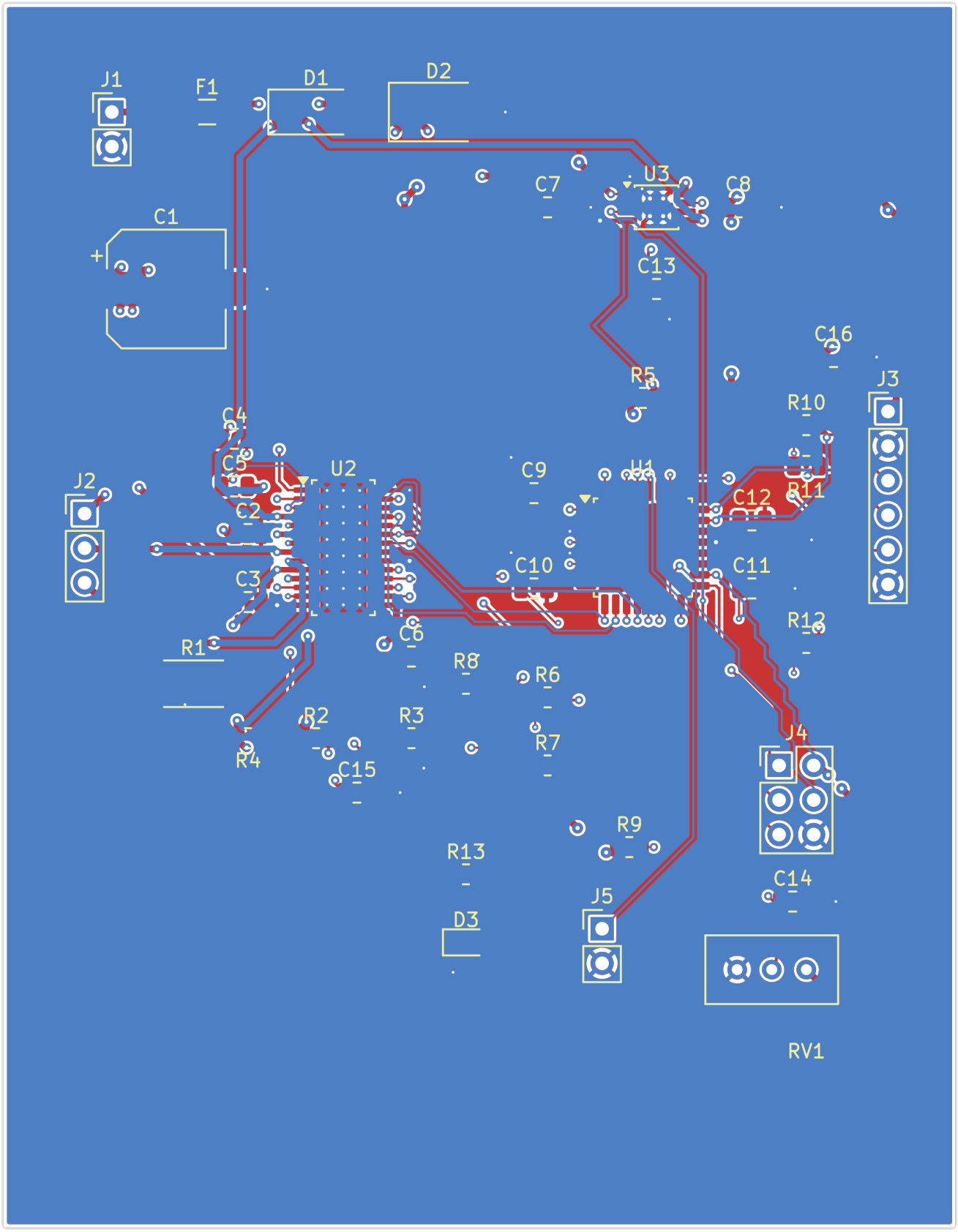

J1
F1
D1
D2
C7
U3
C8
C1
C13
C16
R5
J3
R10
C4
C5
U2
C9
U1
J2
C2
C12
R11
C3
C10
C11
R12
R1
C6
R8
R6
R2
R3
J4
R7
C15
R4
R9
R13
C14
J5
D3
RV1



Copper



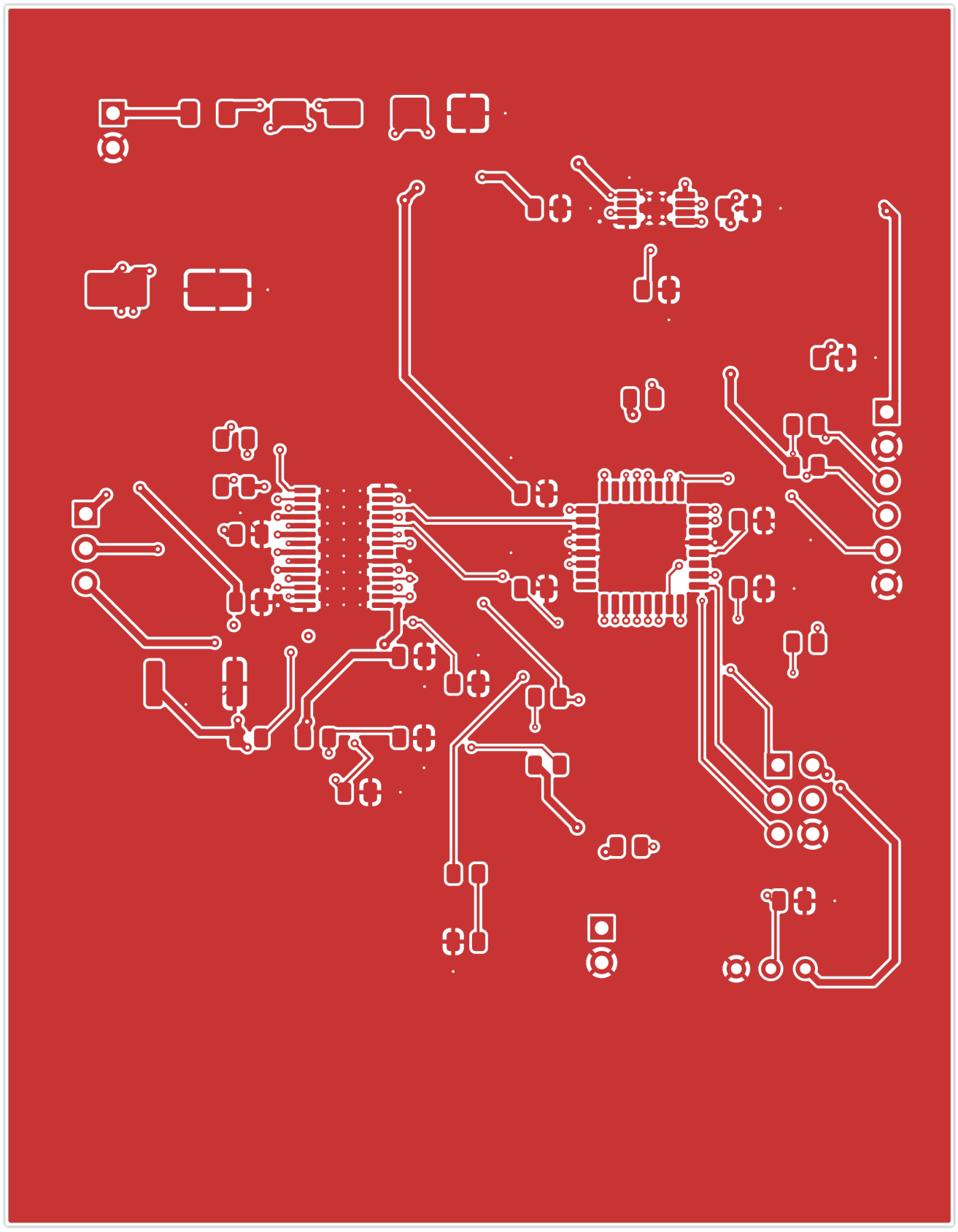
Assembly


J1
F1
D1
D2
C7
U3
C8
C1
C13
C16
R5
J3
R10
C4
C5
U2
C9
U1
C12
R11
J2
C2
C10
C11
C3
R12
C6
R1
R8
R6
R2
R3
R7
J4
R4
C15
R9
R13
C14
J5
D3
RV1

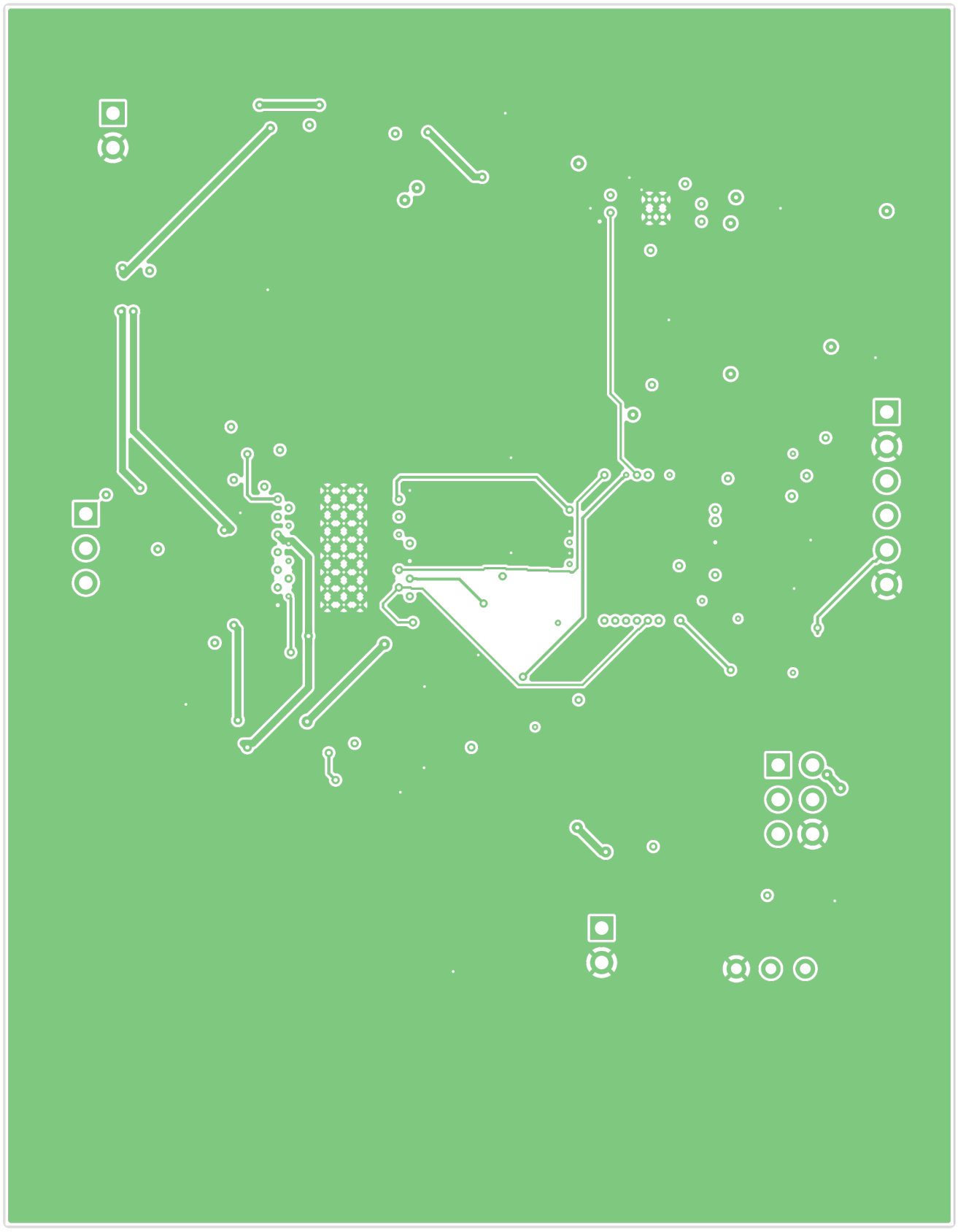
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Copper Layers

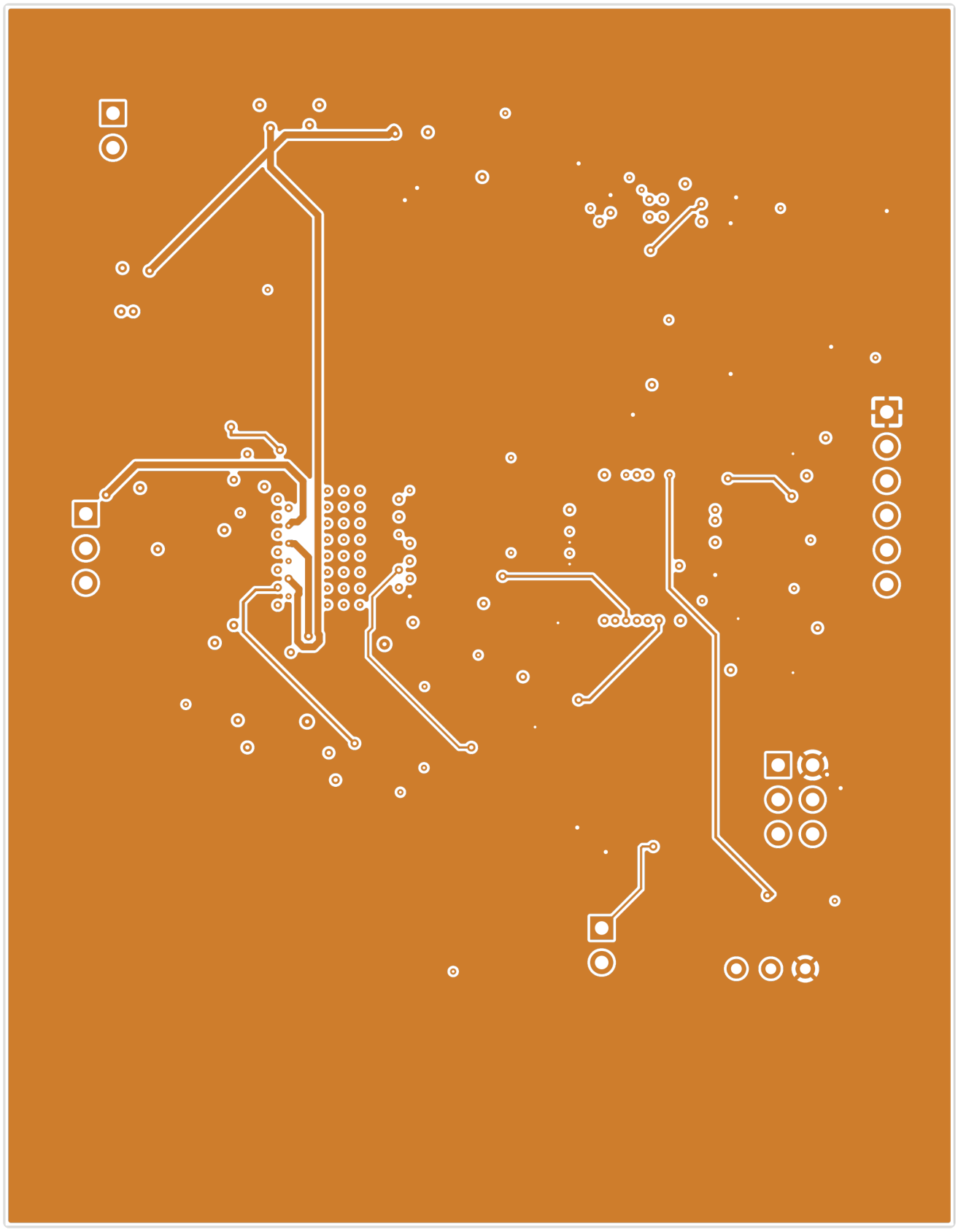
F.Cu



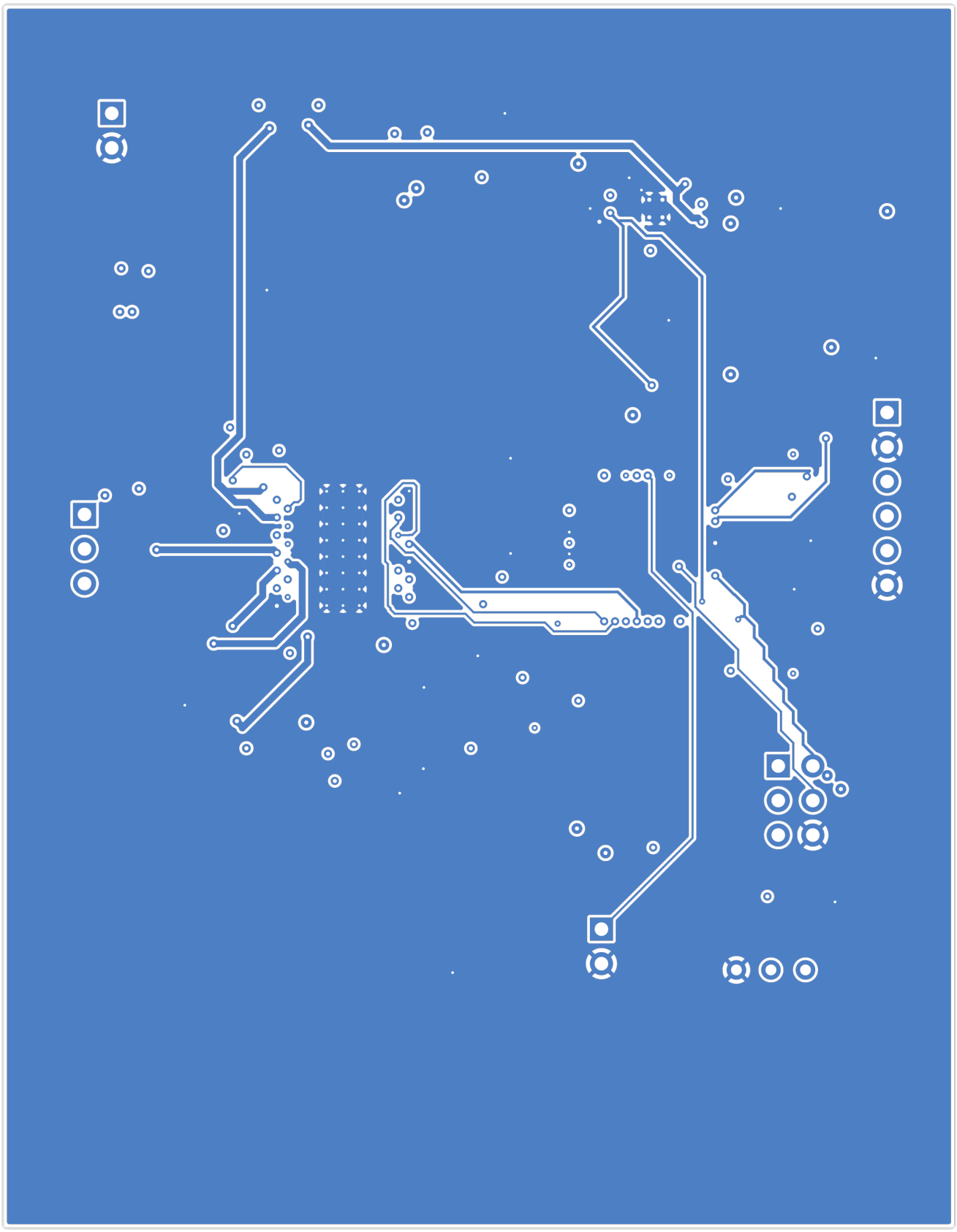
In1.Cu



In2.Cu



B.Cu



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Bill of Materials

Value	Package	Qty	References	MPN	LCSC
100nF	C_0805_2012Metric	10	C2, C3, C5, C9, C10, C11, C12, C14, C15, C16	CC0805KRX7R9BB104	C49678
10nF	C_0805_2012Metric	2	C4, C13	CL21B103KBANNNC	C1710
10uF	C_0805_2012Metric	1	C8	CL21A106KAYNNNE	C15850
1uF	C_0805_2012Metric	1	C7	CL21B105KBFNNNE	C28323
220uF 50V	CP_Elec_8x10.5_PolarityMark	1	C1	EEEFT1H221AP	C178594
470nF	C_0805_2012Metric	1	C6	CL21B474KBFNNNE	C13967
RED	LED_0805_2012Metric	1	D3	NCD0805R1	C84256
SMBJ24A	D_SMB	1	D2	SMBJ24A	C224017
SS36	D_SMA	1	D1	SS36	C16015
1A 63V	Fuse_1206_3216Metric	1	F1	SF-1206S100-2	C3167176
12-24V INPUT	PinHeader_1x02_P2.54mm_Vertical	1	J1	ZX-PZ2.54-1-2PZZ	C7501260
AVR ISP	PinHeader_2x03_P2.54mm_Vertical	1	J4	2.54-2*3P	C65114
HALL	PinHeader_1x06_P2.54mm_Vertical	1	J3	2.54-1*6P	C37208
MOTOR ABC	PinHeader_1x03_P2.54mm_Vertical	1	J2	2.54-1*3P	C49257
RUN SWITCH	PinHeader_1x02_P2.54mm_Vertical	1	J5	ZX-PZ2.54-1-2PZZ	C7501260

Value	Package	Qty	References	MPN	LCSC
0.5R	R_2512_6332Metric	1	R1	WSL2512R5000FEA	C511023
10k	Potentiometer_Bourns_3296W_Vertical	1	RV1	3296W-1-103LF	C34846
10k	R_0805_2012Metric	9	R3, R5, R6, R7, R8, R9, R10, R11, R12	0805W8F1002T5E	C17414
1k	R_0805_2012Metric	2	R4, R13	0805W8F1001T5E	C17513
56k	R_0805_2012Metric	1	R2	0805W8F5602T5E	C17756
ATmega328P-AU	TQFP-32_7x7mm_P0.8mm	1	U1	ATMEGA328P-AU	C14877
DRV8313PWPR	TI_PWP0028C_EP3.4x9.7_Mask3.1x5.18_ThermalVias	1	U2	DRV8313PWPR	C92482
TPS7A1650DGNR	HVSSOP-8-1EP_3x3mm_P0.65mm_EP1.57x1.89mm_ThermalVias	1	U3	TPS7A1650DGNR	C468238

DRC Status (`--mfr jlcpcb` rules)

Metric	Count
Errors	0
Errors: kct (`--mfr jlcpcb` rules)	0
Errors: kicad-cli (geometric)	0
Warnings	0
Blocking	0

Status: PASS

Manufacturing Readiness

Verdict: READY

Action Items

- **[OPTIONAL]** Refill zones to verify 2 zone-connected nets, then re-run kct audit
- **[OPTIONAL]** Confirm mains-named net FUSED is not mains/HV: declared non-HV in the net-class map (12-24 V DC motor supply after F1 (1 A, 63 V fuse), SMBJ24A-clamped; SELV, not mains (issue #8278))
- **[OPTIONAL]** Analog net: GND_SENSE — analog signal; noise-sensitive, avoid crossing digital signals
- **[OPTIONAL]** Analog net: SENSE_IN — analog signal; noise-sensitive, avoid crossing digital signals

Routing Status

Metric	Value
Signal Net Completion	100.0% (35/35)
Overall Completion	100.0%
Complete Nets	37 / 37
Zone-Connected Nets	2
Incomplete Nets	0
Unconnected Pads	0

Zone-Connected Nets

- +5V
- GND

Cost Estimate

Metric	Per Board (estimated)
PCB Fabrication	~3.26 USD
Components (estimated)	~2.23 USD
Assembly (estimated)	~2.19 USD
Total (estimated)	**~7.67 USD**
Batch Quantity	5
Batch Total (estimated)	~38.36 USD